AI LAB WEEK 5

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import random
import math

N = 8
def heuristic(state):
    h = 0

    for i in range(N):
        for j in range(i + 1, N):
            # Same row or same diagonal
            if state[i] == state[j] or abs(state[i] - state[j]) ==
abs(i - j):
                h += 1

    return h

def print_board(state):
    for row in range(N):
        for col in range(N):
            if state[col] == row:
                print("Q", end=" ")
            else:
                print(".", end=" ")
        print()

def get_neighbor(state):
    neighbor = state.copy()

    col = random.randint(0, N - 1)
    new_row = random.randint(0, N - 1)

    while new_row == neighbor[col]:
        new_row = random.randint(0, N - 1)

    neighbor[col] = new_row
    return neighbor

def simulated_annealing():
    # Initial random state
    current = [random.randint(0, N - 1) for _ in range(N)]
    best = current.copy()

    T = 100.0
    cooling_rate = 0.99
    min_temperature = 0.001

    print("Initial state:", current)
    print("Initial cost:", heuristic(current))

    while T > min_temperature:
        current_cost = heuristic(current)
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            if current_cost == 0:
                break
            neighbor = get_neighbor(current)
            neighbor_cost = heuristic(neighbor)

            delta_E = neighbor_cost - current_cost

            if delta_E < 0 or random.random() < math.exp(-delta_E / T):
                current = neighbor
            if heuristic(current) < heuristic(best):
                best = current.copy()
            T *= cooling_rate

        if heuristic(current) < heuristic(best):
            best = current.copy()

        print("\nBest state:", best)
        print("Best cost:", heuristic(best))

        if heuristic(best) == 0:
            print("Solution found!")
        else:
            print("No solution found. Best state is displayed.")

        print("\nFinal chessboard:")
        print_board(best)


simulated_annealing()
```

```
Initial state: [7, 7, 4, 6, 4, 2, 2, 2]
Initial cost: 10

Best state: [2, 0, 5, 7, 1, 3, 6, 2]
Best cost: 1
No solution found. Best state is displayed.

Final chessboard:
. Q . . . . . .
. . . . Q . . .
Q . . . . . . Q
. . . . . Q . .
. . . . . . . .
. . Q . . . . .
. . . . . . Q .
. . . Q . . . .
```